
Alexander Montello

12th Grade

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[LinkedIn](#)

907 Nancy Ln, Cincinnati, OH

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EDUCATION AND WORK EXPERIENCE

Walnut Hills High School

AUG 2018 - MAY 2025 | 4.00 unweighted GPA; 5.44 weighted

Concurrent Enrollment at the University of Cincinnati, Cincinnati State University, and Miami University

MAY 2024 - MAY 2025 | 4.00 unweighted GPAs

Mathnasium, Cincinnati OH

JUNE 2024 - PRESENT | Work part-time teaching specialized math modules to over 30 students K-12.

TECHNICAL SKILLS

Onshape, Autodesk Inventor, Blender, Unity, FDM slicers and CAM software, CNC and manual machining, electronics, Excel and VBA, Android Studio, Visual Studio Code, Python, HTML and CSS, JavaScript, Java, C#, C, Arduino and Raspberry Pi, RStudio, Google Colab, Jupyter Notebooks, Git

Design & Mechanical Engineering

- Used Onshape to design, fabricate, and assemble numerous complex, high-precision robot components, including sensor mounts, motion transmission components, arms, and effectors for three years
- Led mechanical development and data-driven iteration on FIRST Tech Challenge robots for two years while educating team members about mechanical design and component fabrication
- Directed my team's use of tools such as CNC plasma tables and laser cutters to create custom parts, and educated peers on their operation and best practices
- Independently designed, built, wired, and programmed five robotic mechanisms with sensors, integrated circuits, and 3D printing such as an autonomous turret and adjustable gantry
- Participated in two university level summer programs, collaborating on three engineering projects using microcontrollers, sensors, and C code
- Rendered animations of robots and 3D parts using rigidbody and fluid simulations in Blender

Programming

- Created websites for computational tools and a showcase for my own projects with HTML, CSS, and JavaScript
- Implemented Python APIs and libraries including OpenCV, AprilTags, OpenAI, and Matplotlib in individual practice projects, primarily 3D computer vision algorithms
- Simulated new pathfinding algorithms and physics models using Unity and C#
- Contributed to optimizing finite state machine software using Android Studio and Github in robotics
- Researched asteroid danger using rigorous machine learning and statistical models during a Stanford University summer program, presenting data analysis and results
- Collaborated on projects developing, training, and implementing AI (computer vision, LLMs, and classification) for both competitive robotics and a guided internship program

SCORES AND AWARDS

ACT 36 | SAT 1580 | AP 5 on Physics C (Mechanics & Electricity and Magnetism), Calculus BC, Statistics, Computer Science A | Presidential Scholars Program Candidate | National Merit Commended Scholar | 3x FTC Control Award (Regional, State Championship, and World Championship) | 2x FTC Inspire Award (Regional) | 2x FTC Think Award (Regional) | 2x Maxima cum Laude (National Latin Exam) | AP Scholar with Distinction

[Project Portfolio](https://alexmontello.github.io/portfolio) (alexmontello.github.io/portfolio)

